Framework for Blockchain-Based Community-Assisted Privacy- Preserving Reporting Service for Local Governance

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Abstract

Local governments in developing countries often struggle to address public issues e ciently. These struggles are mainly due to limited transparency, centralized data management, and insu cient citizen participation. E-governance and crowd-sourced reporting systems are designed to enhance public engagement. But their e ectiveness is limited by issues related to data integrity, user anonymity, content reliability, and trustworthiness.As a solution to the above challenges, we propose a blockchain-based, community-assisted, privacy-preserving reporting framework for local governance. The main objective of the project is to design and prototype a permissioned blockchain-based reporting system. This system will enable citizens to anonymously submit public issue reports and participate in opinion polling. Also, our system will ensure data immutability and transparency. To manage reporting work ows, issue tracking, and community voting in a secure and tamper-resistant manner, the blockchain will use Proof-of-Authority (PoA) mechanism and smart contracts. User identity is decoupled from the report content using a simulated identity and an access control mechanism to preserve the privacy of the users. Large data objects, such as images, are stored o -chain with cryptographic hashes anchored on the blockchain. Additionally, an Arti cial Intelligence (AI)-assisted content moderation mechanism will be used to minimize the impact of malicious, low-quality reporting content. To demonstrate the feasibility of the proposed framework, A web-based Decentralized App (DApp) is developed as a proof of concept within a controlled experimental environment. Through this project, we aim to deliver a practical and scalable prototype that enhances citizen participation, accountability, and trust in local governance systems.

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Acronyms

AC - Alternating Current
DC - Direct Current
LV - Low Voltage
MV - Medium Voltage

Chapter 1

Introduction

Local governance is fundamental to the upkeep of public infrastructure and the overall well-being of the community. To address public problems such as damaged roads, issues in waste management, street light failures, e ective communication between citizens and local authorities is critical. Recently, the integration of e-governance systems and crowdsourcing has improve the citizen participation and streamline the reporting and resolution of public issues.

However, most of the current systems rely on centralized architectures. This poses a challenge in dealing with data integrity, transparency, and user privacy. Citizens are reluctant to provide complaints about sensitive topics or criticism due to potential retaliation or the possibility of their personal info being abused. At the same time, the local government is faced with the challenge of verifying the authenticity of anonymous complaints raised by citizens.

Some of the key features of Blockchain technology are immutability, transparency, and decentralization. Those features can be used to address several limitations of conventional reporting systems. It is possible to design reporting platforms that protect user anonymity while ensuring the integrity and accountability of reported information by combining blockchain-based architectures and privacy-preserving mechanisms. This project examines the application of blockchain technology in developing a secure, community-driven reporting framework tailored to local governance contexts.

1.1 Problem Background

In the context of local governance, crowdsourced reporting systems and e-governance platforms are widely adopted. By using those systems, citizens can report issues, provide feedback and participate in decision making. It improves the inclusiveness and the accountability of every citizen. In developing countries like Sri Lanka, these systems have the potential to signi cantly improve the governance e ciency.

Timely identi cation and the resolution of public conccerns are the key to governance e ciency.

However, despite their advantages, the conventional e-governance and reporting systems are still highly centralized, making them prone to various threats such as data tampering, illegal access, and single points of failure. In addition, the ability to track behavior down to individual users could serve as a discouraging factor, particularly in reporting delicate matters, in a system that lacks anonymity. Blockchain-based systems have appeared as a feasible solution for managing shared data in a tamper-proof and transparent manner. The ability of permissioned blockchains to enable controlled participation while maintaining accountability through trusted validators makes them a desirable choice for governance applications. By using smart contracts, reporting work ows and opinion polling can be automated rather than relying on centralized intermediaries.

While blockchain technology o ers data integrity and transparency, it still poses challenges related to user privacy, content moderation, and trustworthiness among anonymous participants. These challenges remain open research and implementation problems. Combining blockchain with privacy-preserving identity mechanisms and AI-assisted moderation can help address these limitations and supports the need for a comprehensive framework for secure and reliable community-assisted reporting.

1.2 Problem Statement

The existing frameworks of e-governance and crowdsourced reporting systems for local governance do not provide an integrated mechanism that can guarantee data integrity, user anonymity, and trustworthiness of citizen submitted reports. Centralized architectures are prone to unauthorized access, data tampering, and loss of transparency. At the same time, identity dependent systems discourage citizen engagement due to privacy concerns and fear of retaliation.

Although blockchain has been applied to the context of local government context, many blockchain-based reporting systems still fail to balance anonymity with mechanisms for assessing the reliability and quality of submitted reports. Furthermore, the lack of content moderation and trust evaluation increases the risk of misinformation, and malicious or low quality submissions.

Therefore, there is a clear need for a privacy-preserving, blockchain-based reporting framework that enables anonymous citizen participation while ensuring data immutability, transparent work ows, and reliable evaluation of reported information. This project aims to address this gap by designing and prototyping a permissioned blockchain-based community-assisted reporting system that incorporates smart contracts, o -chain data storage, and AI-assisted content moderation

mechanism suitable for local governance environments.

1.3 Objectives and Scope

1.3.1 Aim

The aim of this project is to design and prototype a blockchain-based, privacy-preserving, community-assisted reporting system for local governance that empowers citizens to report public issues and provide their opinions in a transparent, accountable, and secure manner.

1.3.2 Objectives

1. Design and implement a permissioned reporting blockchain using a PoA consensus mechanism to ensure the integrity and immutability of citizen-submitted reports.

2. Design and deploy smart contracts to automate public issue reporting workflows, including ticket creation, issue resolution and closure, community voting as feedback on resolved issues, and separate opinion polling mechanisms for public consultation.

3. Develop user-friendly interfaces that enable citizens and authorized government officials to interact with the blockchain for reporting issues and tracking their resolution status.

4. Design and integrate an AI-based content moderation system to detect and filter inappropriate textual and visual content in user-submitted reports.

1.3.3 Scope

1. A prototype of the proposed blockchain will be implemented with a minimum of three full nodes. The PoA consensus mechanism will be used to enable efficient and controlled transaction validation within a permissioned network, while excluding deployment of public, permissionless blockchains or globally distributed ledger infrastructures.

2. User authentication and identity decoupling are managed through a simulated identity service, and the design and implementation of a fully decentralized Self-Sovereign Identity (SSI) blockchain are explicitly excluded from the scope of this project.

3. The integration of AI within the system is limited to content moderation using existing machine learning models, libraries, or external Application Programming Interfaces (Application Programming Interface (API)s), and the development of custom, from-scratch deep learning architectures is not within the scope of this work.

4. Transaction validation authority within the blockchain network is restricted to pre-approved institutional stakeholders, including local government entities and recognized non-governmental organizations, and does not permit anonymous public participation or Proof-of-Work (Proof-of-Work (PoW)) based mining.

5. The project emphasizes functional validation, system evaluation, and performance analysis within a controlled experimental environment, and does not address real-world legal compliance, regulatory adoption, legislative reform, or large-scale integration with existing national government information systems.

6. The user interface development is limited to a responsive, web-based DApp accessible via standard mobile and desktop web browsers, and explicitly excludes the development of native mobile applications (iOS or Android) or cross-platform native frameworks.

1.4 Literature Review

1.4.1 Digital Citizen Reporting and e-Governance Systems

Electronic governance (e-governance) is using the information and communication technologies to provide government services. This includes the support of government to citizens (G2C), government to business (G2B) , government to agencies (G2A) etc. By utilizing the e-governance platforms, everyone can gain access to whole range of public services. Additionally citizens can submit complaints and participate in governance decision making process. [?].

Digital citizen reporting system is one of the many use case of e-governance. "FixMyStreet" is a a such existing platform. It enable citizens to report issues for di erent issue categories. Sanitation, Tra c and Public Safety are some examples. Previous studies shows that these kind of systems improve governance administrative e ciency, civic participation and also they increase the transparency.[?].

Even though there are many bene ts of digital citizen reporting in local governance as described earlier, most of them still use centralized server based architecture. This centralized approach have its own limitations and disadvantages.

Susceptibility to data tempering, lack of transparency are the main problems that impact heavily on public trust. In addition to that, mandatory identity disclosure is discouraging public participation on this kind of reporting systems. Therefore these problems motivate the ndings of new decentralized, transparent and immutable architectures that still preserve user privacy with accountability.

1.4.2 Blockchain for Local Governance Applications

Blockchain is a technology that has been popular for its ability to have an immutable, transparent and decentralized ledger for public data. Hence, its a transformative tool for e-governance to address the problems that existing implementations had. The Blockchain allow us to verify the immutability of the records, so its suitable for governance applications where the trust and the accountability is crucial. [?], [?].

Blockchain based land registry system implemented in Cook Country, Illinois is a one real world example of the applicability of blockchain in local governance. They aim to reduce the property fraud by consolidating historical ownership records into a veri able block in the blockchain. Similar to that Delaware has taken initiative to explore the use of smart contracts that's enable real time asset tracking and automate business lings. This improves the regulatory compliance and oversights [?].

In addition to administrative e ciency, blockchain has been utilized for citizen centric services, such as secure digital identity management. Estonia has taken a initiative to integrate blockchain across healthcare, taxation and even electronic voting systems. So this can provide a mathematically veri able audit trail for government data. MyPass project in Austin and the World Food Program's building blocks initiative also use Blockchain technology under the hood to manage encrypted identity and transaction records [?].

Despite the advancements in the blockchain base applications, existing literature highlights the challenges of adoption in governance. Scalability constraints, privacy concerns related to identity data, performance overhead and also the need of the legislative and institutional reforms are some examples. These factors needs to be addressed by carefully selecting blockchain architectures and consensus mechanisms before implementing systems for frequent citizen interactions.

1.4.3 Blockchain-Based Citizen Reporting and Complaint Management Systems

The traditional centralized platforms have data integrity issues, transparency issues and trust issues when it comes to citizen reporting. Past works also indicates that centralized complaint registries often have single point of failures , delayed

responses and public mistrust. Therefore blockchain integrated reporting services can be used to mitigate those issues.

A blockchain based police complaint management system has proposed by Pratheeba [?] contains a modular architecture. It's include security, blockchain, web interface and a hybrid cloud storage. This system ensure immutability of complaint records by cryptographic hash linking. And also it utilize RSA based encryption to restrict the access only to authorized personnel. In Addition to that, Pratheeba used a Natural Language Processing (NLP) techniques to automatically categorize complaints in the system based on their priorities.

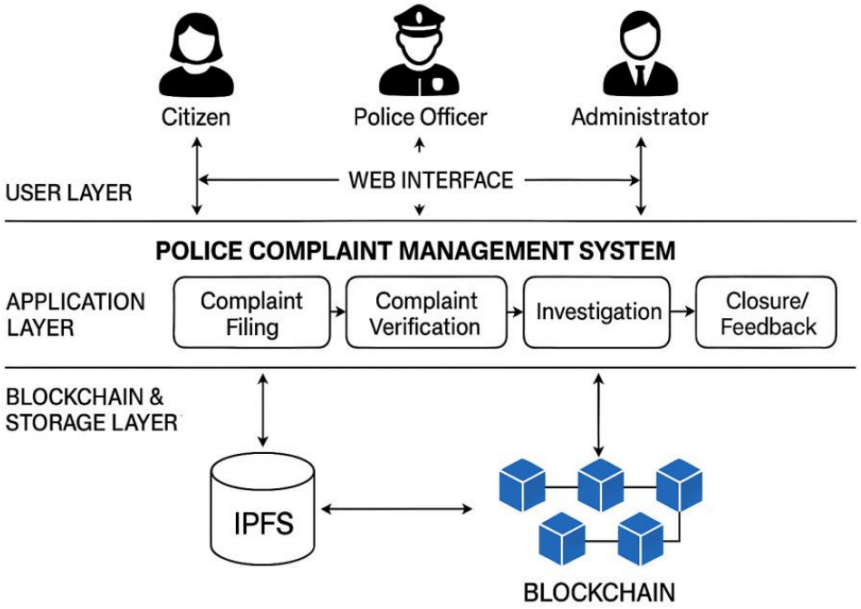

Figure 1.1: System Architecture of the Police Complaint Management System.[?]

A three tier architecture for decentralized complaint management framework has proposed by Manjula [?] is shown in the Figure 1.1. The three layers are web based presentation layer, a business logic layer and nally a blockchain data layer. This system has a role based access control method to involve multiple stakeholders. Citizens, Law enforcement authorities and judicial bodies are some of them. More importantly this system has a immutable storage for multimedia evidences. Intelligent complaint assignment based on workloads, automatic escalation mechanism are some of the key features of the framework. The experimental evaluations shows that a signi cant reductions in complaint processing time. The successful detection of data tampering also archived by blockchain veri cation.

In literature, existing blockchain based complaint management systems shows some improvements in transparency and administrative e ciency. But most of them are domain speci c. They primarily target law enforcement or isolated governance functions. In addition to that several platforms still rely on centralized architectures. Only a limited attention is given for implementing a community assisted validation, for privacy preserving reporting mechanisms and e cient handling of multimedia evidences. Therefore these limitations highlight the need for a generalized reporting framework for local governance.

1.4.4 Permissioned Blockchains and PoA Consensus Mechanisms

Based on the participation and access control blockchain network can be classi- ed in to two categories. [?]. They are permissionless architecture (public) and permissioned architecture. Bitcoin and Early Ethereum networks are permissionless blockchains. They allowed unrestricted participation and relyed on consensus mechanism like PoW or Proof of Stake (Proof-of-Stake (PoS))[?]. These architectures and consensus mechanisms emphasized decentralization among participants. But they often su er from the high latency of the network, limited throughput and unpredictable transaction costs due to congestion. These problems make the traditional permissionless PoW or PoS consensus mechanisms unsuitable for governance and crowdsourced applications[?].

While permissionless blockchains are unsuitable for governance related applications, permissioned blockchains restrict the network participation to authorized entities. This enables more e cient consensus mechanisms and greater control over governance data [?]. This known identities of participants allow permissioned systems to achieve higher performance. And support the regulatory compliances, data privacy and structured governance models as well citemdpi2023unleashing.

Other than PoW and PoS consensus mechanisms, PoA o er signi cant advantages over permissioned blockchains. PoW requires energy intensive computation. PoS relies on the economic stake and introduce wealth centralization risks. But on the other hand PoA depends on the identity and the reputation of pre approved validators[?], [?]. Therefore this PoA consensus mechanism enables high transaction throughput and low latency. And also it mitigate the issues such as the "nothing at stake" problem by o chain legal and reputational accountability [?], [?].

The government systems should have accountability, performance predictability and data con dentiality. The PoA is well suited consensus mechanism for that [?]. By using PoA we can have the ability to identify and regulate validator entities to align blockchain operations. It's because we have the existing legal frameworks that ensure institutional responsibility. Hence, can avoid malicious behavior [?].

In Addition to that PoA avoids the indeterministic transaction fees and support controlled validation environments. This make them viable for handling sensitive governmental data under regulatory constraints[?],[?]. As a result, PoA emerges as a practical consensus mechanism for permissioned blockchain-based reporting systems in local governance contexts.

1.4.5 Smart Contracts for Public Issue Management and Opinion Polling

In governance systems, to improve the transparency, e ciency and citizen participation, smart contracts have been explored as the good mechanism. Smart contracts can automate the governance rules and incentives by immutable code. It can support models like civic intelligence governance (CIG). In CIG, citizen engagement is encouraged through on chain participation and token based incentives[?]. Literature shows that approaches like CIG can improve participation rates and knowledge sharing while reducing relances on centralized intermediaries. In the context of multi organizational governance smart contracts also been used within dual blockchain architectures. This method separates transaction data from contract logic. It balance the transparency and privacy in Government to Government and Business to Business processes[?].

Smart contracts also support business processes in governance by embedding policy rules directly into executable logic. Bid submission, evaluation and contract execution are some of the use cases in the public procurement and tendering processes that smart contract hold its value. It also reduce the oppertunities for corruption and increase the accountability[?], [?], [?]. Similar bene ts are observed in the blockchain based land registry systems. Smart contracts have enable immutable property records automated ownership transfers and enforcement of land use policies. This has reduced fraud and administrative ine ciencies[?], [?].

In the context of voting and opinion polling, smart contracts o er enhanced security, transparency, immutability and veri ability inherited from blockchain[?], [?]. There are advance mechanisms such as Quadratic Voting and also blockchain enabled participatory budgeting. Those mechanisms further extend citizen involvement in decision making. It have expressive preference representation and transparent allocation of public resources[?], [?]. In addition to these advantages, there exists some challenges related to scalability, privacy and system complexity when deploying voting and polling systems to large and diverse populations[?], [?].

Transaction Latency and User Feedback

Blockchain transactions are inherently asynchronous with con rmation times dependent on network conditions and consensus protocols. This latency can lead to confusion among the users because of the system status problems and users can determine those as system failures [2]. The literature recommends to use an optimistic user interface pattern, where user actions are acknowledged immediately while blockchain con rmation occurs in the background.

As advanced design solutions, literature shows progressive status disclosure, distinguishing between o -chain receipt of a report and on-chain veri cation which allows users to track the lifecycle of an issue without being exposed to unnecessary low level blockchain details [7], [8]. Mechanisms like that are particularly relevant for issue reporting systems that require transparency and trust.

Transaction Costs and Abstraction of Blockchain Complexity

A signi cant usability challenge coming from the blockchain is transaction cost that associated with public blockchains. And also the concepts of gas fees and uctuating gas costs are very unfamiliar in public services context. Most of the time users expect free access to the reporting platforms[1]. The literature strongly suggest that the use of gas fee abstraction techniques. Meta Transactions is one of the example. In Meta, backend relayer services submit transactions on behalf of the user. Its achieved by preserving cryptographic intent veri cation[2]. Therefore this approach enable citizen to interact with blockchain without technical or economical complexities.

Comparison with Centralized Reporting Platforms

FixMyStreet and SeeClickFIx are existing centralized platforms. Eventhough they are centralized, they provide a valuable benchmarks for crowdsource citizen reporting interfaces. Inituitive map based reporting, simple authentication mechanisms and clear feedback loops are the key characteristics that attributed to platform success[9], [10].

With addition to that SeeClickFix further uses a gami cation mechanism to encourage the users to participate in the system. In contrast, Even though the proposed application uses blockchain for reporting platform, it must retain the usability and the expectations of the users. The literature highlights the need of interface design such that gurantees the data integrity and auditability using familiar ui elements rather than cryptographic representations[11], [8], [12].

Accessibility and Inclusivity Considerations

Inclusivity is a fundamental requirement for e-governance systems. According to [13], it can be seen that poorly designed blockchain interfaces may digitally divide users. Also it may favor crypto-literate users excluding vulnerable populations. To reduce this issue, literature suggests mobile rst design approaches. Furthermore, lightweight client architectures, e ecient bandwidth synchronization can used to support access in resource limited environments. [14].

1.4.7 AI-Based Content Moderation in Civic Reporting

Since blockchain is inheritably immutable its introduce a unique challenge for content moderation in a civic reporting system. In a centralized platforms, administrators can remove or edit inappropriate content. But in a decentralized blockchain based system data cannot be altered once con rmed [15]. While immutability strengthens transparency and accountability, it also raises the risk of permanently storing harmful, illegal, or misleading content. Therefore the literature focuses the need for pre moderation strategies to prevent unsuitable content from being recorded on chain or to limit its visibility after publication [7].

1.4.8 Need for Automated Content Moderation

Civic reporting platforms are susceptible to several forms of misuse that can compromise system reliability and public trust. Automated spam submissions, the submission of malicious or legally sensitive content, and deliberate dissemination of false or misleading reports are some of the common issues [16], [17], [18]. Manual moderation is impractical for above situations. Since the reporting system is large latency, human bias, and operational costs can occur. Furthermore it reintroduces the centralized control structure that blockchain system is aim to minimize.

Therefore the literature supports AI assisted moderation as a scalable and e cient approach for ltering content before its permanently recorded. It maintain the acceptable levels of decentralization [19], [20].

1.4.9 Architectural Approaches for AI Blockchain Integration

Due to the computational and storage limitations of blockchain platforms, AI-based moderation cannot be performed directly on chain. O chain processing combined with on chain veri cation identi ed by existing researches as most practical approach[20], [21].

Chapter 2

Background

2.1 Blockchain Technology

Blockchain is a distributed, digital, append-only ledger technology. It enables data to be stored in immutable and transparent manner across multiple nodes in a network. Unlike centralized systems, blockchain systems maintain a syncronized copy of the ledger among all authorized participants. This approach reduces the dependency on a single intermediary controlling authority. Each block in the blockchain contains a set of validated transactions, a timestamp, and a cryptographic hash of the previous block. This hash-chaining mechanism ensures that any alteration to previous recorded data becomes immediately detectable.

Immutability is a key characteristic in blockchain technology. It is computationally impractical to alter or delete a transaction after it is con rmed and appened to the blockchain. The immutable property is signi cantly important in governance-related systems, because integrity and transparency are a must.

Cryptography plays a signi cant role in blockchain systems. To generate unique digital ngerprints of data, hash functions are used. To verify ownership and authenticity of transactions, digital signatures are used. By using above cryptographic methods, blockchain systems gurantee integirty, authenticity and non-repudiation of data.

2.2 Permissioned Blockchain and PoA

Bitcoin and Ethereum are permissionless blockchains, where anyone can participate as a node and validate transactions. In contrast, permissioned blockchains restrict participation to a prede ned set of entities. These entities, often referred to as validator nodes. They are responsible for validating transactions and maintaining the integrity of the blockchain.

While permissionless blockchains rely on consensus mechanisms like Proof of Work or Proof of Stake, the Proof of Authority (PoA) consensus mechanism is specially designed for permissioned blockchains. In here instead of relying on computationally expensive mining or staking process the validator nodes in the permissioned blockchain are validating blocks. Therefore this approach is a reputation and indetity-based mechanism, where the authority of the validators is based on their identity and reputation within the network. This allows for faster transaction processing and lower energy consumption compared to traditional consensus mechanisms, making it suitable for enterprise applications and private networks.

In the context of local governance perspective, such system requires predictable performance, immutability, transparency and accountability. PoA consensus running in a permissioned blockchain is well suited for this requirements because validators identities are known and they controlled by trusted entities. With addition to that, PoA eliminates the need for economic incentives. This make it more e cient and practical for local governance applications.

2.3 Smart Contracts

Smart contracts are rst introduced by Ethereum blockchain. They are self-executing contracts with the terms of the agreement directly written into code. In a blockchain based governance systems smart contract can be used to eliminate centralized administrative control. It can automate work ows that would traditionally require human intervention. Therefore this reduce the reliance on intermediaries, hence improve the transparancy and trusthworthiness.

The proposed system uses Solidity based smart contracts deployed on a private permissioned PoA blockchain.

2.4 Decentralized Applications (DApps)

Decentralized Applications (DApps) are software applications that operate on a decentralized network, such as a blockchain, rather than being hosted on centralized servers. Unlike traditional web applications that rely on centralized databases and backend infrastructure, DApps utilize smart contracts to handle backend logic and state management. This shift from traditional Web2 to Web3 architecture ensures that application data is immutable, transparent, and resistant to single points of failure or centralized control.

In a typical DApp architecture, the frontend provides the user interface, while the core logic is distributed across the blockchain network. To interact with the blockchain, the frontend utilizes Web3 provider libraries, such as Web3.js or Ethers.js. These libraries enable the application to communicate with blockchain

nodes via Remote Procedure Calls (RPC). Through this connection, the frontend can query the blockchain to read the current state of smart contracts or construct new transactions. When a user wishes to modify the blockchain state, they must cryptographically sign the transaction using their digital wallet's private key, ensuring authenticity and authorization.

A signi cant challenge in traditional DApp adoption is the requirement for users to pay transaction fees, commonly known as "gas," using the native cryptocurrency of the network. This creates a barrier to entry for regular users who may not possess or understand cryptocurrency. To address this usability issue, modern DApps employ meta-transactions and relayers. A meta-transaction allows a user to cryptographically sign their intent to execute a transaction o -chain without broadcasting it themselves. This signed message is then sent to an intermediary service known as a relayer. The relayer, which holds a balance of the network's cryptocurrency, wraps the user's signed message in a standard blockchain transaction, submits it to the network on behalf of the user, and pays the associated gas fees. This approach provides a seamless, "gasless" user experience while maintaining the security and non-repudiation guarantees of cryptographic signatures, making blockchain applications more accessible to the general public.

2.5 InterPlanetary File System (IPFS)

The InterPlanetary File System (IPFS) is a peer-to-peer network protocol designed for storing and sharing les in a decentralized manner. Unlike the traditional web, where les are identi ed and accessed through server-based addresses such as URLs, IPFS locates les based on their content. This content-addressed approach improves resilience, reduces dependence on centralized servers, and provides an e cient o -chain storage mechanism for blockchain-based applications. A fundamental concept in IPFS is the Content Identi er (CID), which is a cryptographic hash generated from the content of a le. If the le content changes, the CID also changes, ensuring strong integrity guarantees because users can verify that the retrieved content exactly matches the expected CID. Since identical content generates the same CID, IPFS also supports deduplication and immutable references. In the IPFS network, data can be hosted by any node that chooses to store or pin the content, allowing les to be retrieved from multiple peers using their CID. This decentralized storage model increases availability, improves fault tolerance, and reduces reliance on a single server, although long-term persistence depends on nodes continuing to pin the data. In blockchain-based systems, IPFS is commonly used as an o -chain storage layer because storing large les directly on-chain is costly and ine cient. Instead, applications store only CIDs and essential metadata on the blockchain, while the actual content remains in IPFS. Users can then retrieve

1. Cryptographic Veri cation: The relayer receives the payload via a POST /report endpoint. Utilizing the ethers.js library, it performs a strict dual-signature validation:

   (a) It veri es the ZKP Govid Simulator ticket signature against the dynamically fetched Government Authority public key to ensure the user is an authorized citizen.

   (b) It reconstructs the payload hash (binding the description and ticket ID) and veri es the citizen's signature to prevent data tampering in transit.

2. AI Content Moderation: To prevent on-chain spam, toxic immutability of blockchains and ensure platform integrity, the relayer routes the submitted text and media to the AI Oracle moderation service. This service acts as an automated lter, analyzing the content for appropriateness and returning a validation verdict before any storage operations occur.

3. Decentralized Storage (IPFS): To maintain a highly e cient and cost-e ective blockchain state, heavy multimedia les and detailed text are o oaded to the InterPlanetary File System (IPFS). The relayer manages the le bu ers via multer, processes the upload, and retrieves an immutable Content Identi er (CID).

4. Gasless Blockchain Submission: As the nal step, the relayer formats a transaction containing the IPFS CID and the ZKP ticket (which serves as a Sybil-resistant submissionNulli er). The relayer signs and dispatches this transaction to the Reporting.sol smart contract. By paying the network gas fees from its own wallet, the relayer enables a completely frictionless, gasless experience for the end-user.

3.2.5 IPFS O -chain Storage Integration

File Upload Work ow CID Retrieval Integrity Veri cation

3.2.6 AI-based Moderation System Implementation

Balanced coverage only.

Subsections:

Oracle Architecture Oracle Workers Aggregation Logic Moderation Work ow Security Mechanisms

Do NOT over-expand this section.

and smart contracts to communicate with ledger enabling functions such as report submission, transaction querying and state synchronization.

3.2.2 Smart Contract Implementation

Report Contract Ticket Validation State Transition Logic

3.2.3 Identity and Access Control Implementation

To satisfy the project's requirement for privacy-preserving accountability, an identity decoupling mechanism was successfully developed. This subsystem ensures that citizens can interact with the blockchain anonymously while remaining verified by an o -chain authority. The implementation was structured into three main components

- ˆ GovID Simulator: An o -chain simulated identity server was developed and deployed to a personal Virtual Private Server (VPS). This service acts as the trusted authority for verifying real-world identities. A dedicated API endpoint (/api/govid/verify-citizen) was implemented to securely authenticate users via their National Identity Number (govId) and a password, bridging the gap between traditional web authentication and decentralized access.
- ˆ Citizen Keypair generation: To completely decouple a user's real identity from their on-chain footprint, the identity server does not store or transmit blockchain addresses. Instead, upon successful authentication, the simulator generates and returns a secure, 256-bit citizenSeed. This cryptographic seed is then used by the client-side DApp to deterministically generate an anonymous public-private keypair, allowing the citizen to interact securely with smart contracts without exposing personal data.
- ˆ Ticket Batch Generation: To facilitate multiple secure, anonymous report submissions over time, the simulator implements a batch authorization protocol. During authentication, the server generates a structured ticketBatch (currently con gured to issue 10 tickets per request). Each ticket contains a unique ticketId (a cryptographic hash) and a signature signed by the Identity Authority's private key. Citizens use these single-use tickets to authorize their transactions on-chain. The smart contract validates the authority's signature on the ticket, granting access without ever knowing which speci c citizen is submitting the report.

signature.

To ensure integrity of the moderation results, several mechanisms are used. The aggregator computes a hash of the submitted report payload. The nal moderation result and oracle votes are combined and hashed to crate the decision hash. Also, the aggregator sign the decision hash using its private key.

3.2.7 Web-based DApp Development

The decentralized application (DApp) serves as the primary gateway for citizens to interact with the local governance system. It is built using a modern web stack comprising Next.js and React for the frontend interface, Tailwind CSS for responsive styling, and Ethers.js for cryptographic operations. The architecture is designed to abstract blockchain complexities away from the end-user while maintaining strict privacy and security guarantees.

User Authentication Interface

The authentication interface is designed to verify citizen eligibility without compromising personal privacy.

- GovID Veri cation Integration: The interface securely collects the citizen's GovID credentials and interfaces with an external GovID Simulator API. This abstracts the process of verifying a citizen's real-world identity against a centralized registry.
- Cryptographic Credential Generation: Upon successful identity veri - cation, the system retrieves a batch of Zero-Knowledge Proof (ZKP) tickets and a unique `citizenSeed`.
- Local State Management: Using the `ethers.js` library, a localized Citizen Wallet is derived from the seed. A custom React Context (CitizenContext) is utilized to manage the session state, ensuring that the generated private keys and ZKP tickets remain strictly on the client-side device and are never exposed to external servers. The UI incorporates dynamic feedback mechanisms to indicate the generation of cryptographic proofs to the user.

Report Submission Interface

The report submission module allows authenticated citizens to report civic issues anonymously.

- Form Design and Data Handling: Developed with a mobile- rst approach, the form captures issue descriptions and optional photographic evidence using standard FormData structures to support multipart uploads.

- Anonymous Cryptographic Signing: To guarantee anonymity and prevent spam, each submission consumes a single ZKP ticket from the user's allocated batch. The payload (consisting of the issue description and the ZKP ticket ID) is hashed using Keccak-256 and subsequently signed using the citizen's locally stored ECDSA private key.

- Meta-Transaction Relaying: Instead of submitting transactions directly to the blockchain and incurring gas fees, the frontend dispatches the signed payload, image, and ZKP metadata to a NestJS Backend Relayer. The relayer handles the intermediate steps: uploading the evidence to IPFS, requesting AI moderation, and finally submitting the transaction to the Geth blockchain network on behalf of the user.

Blockchain Query Interface

The query interface acts as a community feed where citizens can view, filter, and track the status of reported issues.

- Mobile-First Feed Architecture: Implemented as a single-column, scrollable interface inspired by modern social media platforms (e.g., Reddit). This design choice prioritizes readability and user engagement on mobile devices.

- Dynamic Issue Cards: Data retrieved from the blockchain is formatted into interactive cards. Each card displays critical metadata including categorical status badges (e.g., PENDING VALIDATION, OPEN, RESOLVED), community upvotes, truncated descriptions, timestamps, and image thumbnails.

- State Management and Navigation: The interface employs React state hooks to allow real-time filtering of issues by category (e.g., Infrastructure, Parks & Rec, Safety). Furthermore, a persistent Floating Action Button (FAB) is implemented to provide users with seamless access to the report submission workflow at any time.

Chapter 4

Testing / Experimentation / Validation / Veri cation

This chapter MUST look research-oriented.

4.1 Experimental Setup

Describe:

VPS specs, blockchain environment, Docker setup, software versions, datasets.

4.2 Testing Methodology

Explain:

subsystem testing, integration testing, functional testing.

4.3 Blockchain Performance Testing

Metrics:

latency, throughput, block con rmation.

4.4 Smart Contract Validation

Test:

report creation, state transitions, ticket validation.

4.5 AI Moderation Testing

Metrics:
moderation response time, classi cation correctness, voting consistency.

4.6 IPFS Storage Testing

To evaluate the e ectiveness and reliability of the IPFS o -chain storage integration, several testing and validation procedures were conducted focusing on upload performance, retrieval integrity, and CID validation. The upload time was measured for les of di erent sizes in order to assess the responsiveness of the system and the overhead associated with decentralized storage. During each test, the start and end times of the upload process were recorded, and the total duration was calculated. The results indicated that smaller les were uploaded quickly, while larger les required proportionally more time due to the distributed nature of the IPFS network. Retrieval integrity tests were performed to con rm that les downloaded from IPFS remained identical to the originals. After uploading a le and obtaining its Content Identi er (CID), the same le was retrieved from IPFS using the CID and compared byte-for-byte with the original le. The successful matches demonstrated that IPFS reliably preserves le integrity during both storage and retrieval operations. In addition, CID validation tests were conducted to ensure that the generated CID accurately represented the corresponding le content. For each uploaded le, the CID produced by IPFS was recorded, and after retrieval, the CID was recalculated from the downloaded le and compared against the original value. Matching CIDs con rmed that the content had not been modi ed and veri ed the integrity guarantees provided by the IPFS content-addressing mechanism. Overall, these tests demonstrate that the implemented IPFS integration satis es key requirements for decentralized storage, including e cient le uploads, reliable retrieval, and strong veri cation through cryptographic content identi ers.

4.7 End-to-End Integration Testing

To ensure the seamless operation of the entire platform, end-to-end integration testing was conducted by simulating the complete lifecycle of a civic issue report. This testing phase veri es the interoperability between the frontend DApp, the backend relayer, the IPFS storage network, the AI moderation oracle, and the underlying permissioned blockchain.

The end-to-end testing work ow was executed as follows:

1. Citizen Authentication and Key Generation: The test commenced

Figure 4.2: Report Submission Form

Appendix A

Smart Contract Code

Appendix B

Oracle API Endpoints

Appendix C

Docker Compose Files

Appendix D

System Diagrams

Appendix E

Testing Screenshots

Appendix F

GitHub Repository Links

Bibliography

[1] MDPI, \The development of user-centric design guidelines for web3 applications: An empirical study," 2025. Accessed on January 8, 2026.

[2] N. Sonule, \User experience (ux) in web3 and dapps: Challenges and opportunities," 2025. Accessed on January 8, 2026.

[3] Ramotion, \Ui/ux design for web 3.0," 2026. Accessed on January 8, 2026.

[4] F. Weekly, \Why usability is the missing layer in crypto nance," 2025. Accessed on January 8, 2026.

[5] O. Academic, \Governance and societal impact of blockchain-based self-sovereign identities," 2022. Accessed on January 8, 2026.

[6] Fraunhofer, \Conducting a usability evaluation of decentralized identity management solutions," 2026. Accessed on January 8, 2026.

[7] DZone, \Blockchain + ai integration: The architecture nobody's talking about," 2026. Accessed on January 8, 2026.

[8] C. V. Marian, D. A. Mitrea, D. S. Rusu, and A. Vasilateanu, \Transparent digital governance: A blockchain-based work ow audit application," 2025. Accessed on January 9, 2026.

[9] S. F. King and P. Brown, \Fix my street or else: Using the internet to voice local public service concerns," 2012. Accessed on January 8, 2026.

[10] mySociety Research, \Fixmystreet!," 2026. Accessed on January 9, 2026.

[11] O. Adeyeye, \BLOCKCHAIN AND THE FUTURE OF E-GOVERNMENT: STRATEGIC IMPLICATIONS FOR POLICYMAKERS," 2025. Accessed on January 9, 2026.

[12] S. Nikonenko, \Blockchain UX explained: Challenges, trends, and real solutions," 2025. Accessed on January 9, 2026.

[13] A. Bertrand and E. G. Consulting, \How can digital government connect citizens without leaving the disconnected behind?," tech. rep., Ernst & Young (EY), June 2023. EY Connected Citizens Report. Accessed on January 9, 2026.

[14] N. Shahin and L. Ismail, \Towards trustworthy sign language translation system: A privacy-preserving edge{cloud{blockchain approach," Mathematics, vol. 13, no. 23, p. 3759, 2025. Accessed on January 9, 2026.

[15] Tencent Cloud, \How can content moderation use blockchain technology?," 2025. Accessed on January 9, 2026.

[16] M. A. Wani, M. ElA endi, and K. A. Shakil, \AI-generated spam review detection framework with deep learning algorithms and natural language processing," Computers, vol. 13, no. 10, p. 264, 2024. Accessed on January 9, 2026.

[17] G. Gosztonyi, D. Gyetvan, and A. Kovacs, \Theory and practice of social media's content moderation by arti cial intelligence in light of european union's AI act and digital services act," Politics in Central Europe, 2024. Accessed on January 9, 2026.

[18] V. Kolluri, T. R. Gatla, and S. T. Boppiniti, \AI-powered citizen science: Crowd-learning, data veri cation, and public engagement," Scienti c Endeavours, vol. 1, no. 1, pp. 53{60, 2025. Accessed on January 9, 2026.

[19] S. Mohammadi and T. Yasseri, \AI feedback enhances community-based content moderation through engagement with counterarguments," October 6 2025. arXiv preprint arXiv:2507.08110v3. Accessed on January 9, 2026.

[20] S. Goundar and I. Gondal, \AI-blockchain integration for real-time cybersecurity: System design and evaluation," Cybersecurity and Privacy, vol. 5, p. 59, August 14 2025. Accessed on January 9, 2026.

[21] N. Kader, I. Kang, and O. Seneviratne, \Enhancing web spam detection through a blockchain-enabled crowdsourcing mechanism," October 1 2024. arXiv preprint arXiv:2410.00860v1. Accessed on January 9, 2026.

[22] J. Li, L. Jiang, H. Liang, T. Peng, S. Wang, and H. Wei, \Block-CITE: A blockchain-based crowdsourcing interactive trust evaluation," Applied System Innovation, vol. 6, p. 245, October 1 2025. Accessed on January 9, 2026.

[23] Kava News, \AI-powered oracles: Bridging blockchains with smarter data," May 29 2024. Accessed on January 9, 2026.

[24] K. Zintus-art, B. Vass, and J. Ward, \Empirical evidence in AI oracle development," March 21 2025. Chainlink Blog. Accessed on January 9, 2026.

[25] CodeZeros, \AI-powered oracles: How generative AI is enhancing blockchain data feeds," April 10 2025. Accessed on January 9, 2026.

[26] M. Prasad, \Blockchain + AI in combating deepfake content circulation," Scienti c Journal of Arti cial Intelligence and Blockchain Technologies, April 2024. Accessed on January 9, 2026.

[27] A. Kumar, \Hybrid moderation models: Balancing AI and human oversight," October 27 2025. Infosys BPM. Accessed on January 9, 2026.

[28] B. Kae, \Research: AI agent sector overview," March 10 2025. The Block Research. Accessed on January 9, 2026.

[29] Chase Advisors, \Ai and content moderation," tech. rep., Chase Advisors, January 2024. Accessed on January 9, 2026.

[30] K. Mounika and N. R. Reddy, \An integrated machine learning framework for spammer and fake user detection in online social networks," Fringe Management, Economics and Politics, 2024. Accessed on January 9, 2026.